

Sophia Beebe – CV

Sophia Beebe

Software Engineering Student | Robotics & Data Analytics | Project Manager

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Currently

Undergraduate Software Engineering student at Brigham Young University of Idaho, specializing in applied robotics, data analytics, and software architecture, with hands-on experience building full-stack and hardware-integrated systems.

Specialized in

Software Engineering, Robotics Automation, Data Analytics, Project Management, Design Patterns, Embedded Systems, Python, C#, SwiftUI

Education

2021 - Present

Brigham Young University–Idaho

Bachelor of Science, Software Engineering

- Minor: Data Analytics
- Minor: Intercultural Studies
- President, Interdisciplinary Studies Club

Projects

2024 – 2025

Folding Clothes Robot (Capstone Project)

- Designed and built a multi-phase robotic system capable of folding garments
- Implemented servo and stepper motor control using Arduino and Python
- Integrated mechanical design, electronics, and software architecture
- Explored real-world constraints such as torque limits, safety, and repeatability
- Developed two robotic approaches and evaluated tradeoffs between them

2025

Sun Buddy (iOS Application Prototype)

- Built a SwiftUI prototype for tracking daily sunlight exposure
- Designed custom UI components including progress rings and dashboards
- Focused on wellness, behavioral design, and data visualization

2025

Hackathon Project – Intelligent Task & Routine Assistant

- Collaborated on a rapid-development software project focused on improving executive function and daily task management
- Designed and implemented core features under a strict time constraint, prioritizing usability and functional scope
- Applied software engineering principles to break down requirements, assign tasks, and deliver a working prototype
- Integrated data handling and logic to support task tracking, prioritization, and user interaction
- Presented the final product with a live demo, explaining technical decisions and future scalability

Experience

2023 – Present

Teaching Assistant, BYU–Idaho

- Assisted students with HTML, CSS, Python, R, GitHub workflows, and debugging
- Led in-class demonstrations and one-on-one mentoring
- Supported grading, rubrics, and project evaluations

2024

Project Manager (Academic)

- Elected as Project Manager for a 21-person software engineering class project
- Led the end-to-end planning and execution of a large-scale ride-share application
- Authored a 200+ page architectural design document detailing system architecture, data flow, APIs, security, and scalability considerations
- Coordinated cross-functional teams, resolved scope and timeline challenges, and maintained alignment with academic and technical requirements
- Delivered the project at the highest quality level, earning the highest grade ever awarded in the course

Skills

- **Languages:** Python, C#, R, Swift, SQL, HTML/CSS
- **Tools:** Git/GitHub, Arduino, Firebase, ROS concepts, Xcode, RStudio
- **Concepts:** Software architecture, data analysis, robotics control, UI design
- **Soft Skills:** Technical communication, mentoring, leadership, project planning

Awards & Leadership

- President, Interdisciplinary Studies Club

- Certified Yoga Teacher
- Academic Project Leadership Recognition